Answer ALL Questions (answer Q1 and Q2 in one booklet and Q3 and Q4 in a separate one)

Question 1 on Flexible AC Transmission System (FACTS)

1. A long transmission line interconnects two large AC power systems. The voltages at both ends of the transmission line are maintained at 1 p.u. The reactance of the transmission line is 0.4 p.u. and its resistance and shunt charging can be neglected. The system operator is planning to install shunt reactive compensation at the midpoint of the transmission line to regulate the midpoint voltage at 1 p.u.. For this, the following three options are being considered: i) an SVC with a rated susceptance of 2 p.u. ii) an SVC with a rated susceptance of 3.5 p.u. and iii) a STATCOM with a current rating of 3.2 p.u.

   a) Calculate the increase in steady-state stability limit (with respect to the uncompensated case) that can be achieved with each of the three shunt reactive compensation option stated above. Sketch the power-angle curve for each case and label the salient points.

[15]

i) Option 1: SVC with a rated susceptance of 2 p.u.

$B_{max1} = 2.0$ p.u.

When the reactive capacity of the SVC is reached

$$\frac{4V^2}{X}\left(1-\cos\frac{\delta_{lim1}}{2}\right) = V^2 B_{max1} \rightarrow \delta_{lim1} = 2\times\cos^{-1}\left(1-\frac{XB_{max1}}{4}\right) = 73.74°$$

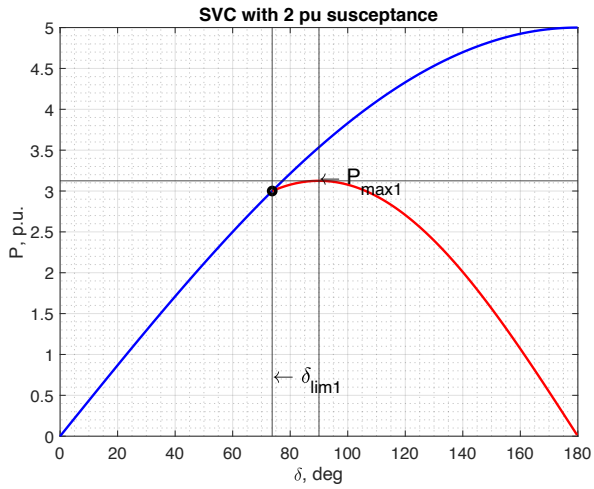
[1 mark]

As $\delta_{lim} < 90°$ the maximum power that can be transferred considering the reactive capacity of the SVC is

$$P_{max1} = \frac{4V^2}{X(4-XB_{max1})} = 3.125 \text{ p.u.}$$

[1 mark]

Model Answers



[1 mark]

Without any compensation

$$P_{max0} = \frac{V^2}{X} = 2.5 \text{ p.u.}$$

Increase in steady-state stability limit
$\Delta P_{max1} = P_{max1} - P_{max0} = 0.625$ p.u.
[1 mark]

ii) Option 2: SVC with a rated susceptance of 3.5 p.u.

$B_{max2} = 3.5$ p.u.

When the reactive capacity of the SVC is reached

$$\frac{4V^2}{X}\left(1 - \cos\frac{\delta_{lim2}}{2}\right) = V^2 B_{max2} \rightarrow \delta_{lim2} = 2 \times \cos^{-1}\left(1 - \frac{XB_{max2}}{4}\right) = 98.917°$$

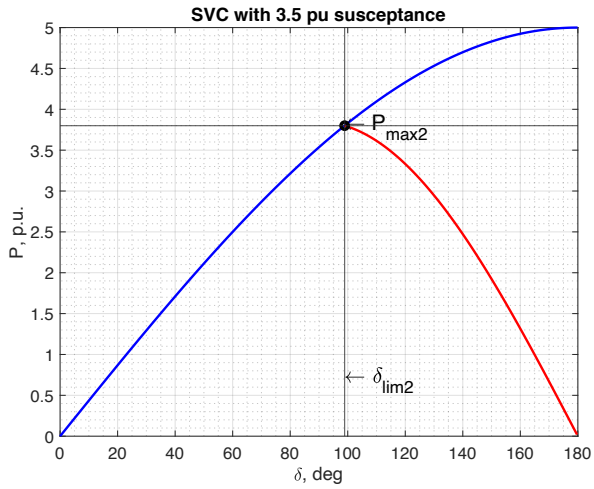
[1 mark]

As $\delta_{lim} > 90°$ the maximum power that can be transferred considering the reactive capacity of the SVC is

$$P_{max2} = \frac{2V^2}{X} \times \sin\left(\frac{\delta_{lim2}}{2}\right) = 3.7997 \text{ p.u.}$$

[1 mark]

Model Answers



[1 mark]

Increase in steady-state stability limit
$\Delta P_{max2} = P_{max2} - P_{max0} = 1.2997$ p.u.

[1 mark]

iii) Option 3: STATCOM with a rated current of 3.2 p.u.

$I_{max} = 3.2$ p.u.

When the rated current capacity of the STATCOM is reached

$$\frac{4V^2}{X}\left(1-\cos\frac{\delta_{lim3}}{2}\right) = VI_{max} \rightarrow \delta_{lim3} = 2\times\cos^{-1}\left(1-\frac{XI_{max}}{4V}\right) = 94.31°$$

[1 mark]

Beyond this point

$$P = \frac{V^2}{X}\sin\delta + \frac{VI_{max}}{2}\sin\frac{\delta}{2} = P_{max0}\sin\delta + \frac{I_{max}}{2}\sin\frac{\delta}{2}$$

as $V = 1.0$ p.u.

[1 mark]

Power angle at which maximum power is transferred can be determined as follows:

$$\frac{dP}{d\delta} = 0 \rightarrow P_{max0}\cos\delta + \frac{I_{max}}{4}\cos\frac{\delta}{2} = 0 \rightarrow 2P_{max0}\left(\cos\frac{\delta}{2}\right)^2 + \frac{I_{max}}{4}\cos\frac{\delta}{2} - P_{max0} = 0$$

[1 mark]

Model Answers

$$\delta_{Pmax} = 2 \times \cos^{-1}\left[\frac{-\frac{I_{max}}{4} \pm \sqrt{\left(\frac{I_{max}}{4}\right)^2 + 8{P_{max0}}^2}}{4P_{max0}}\right] = 101.66°$$

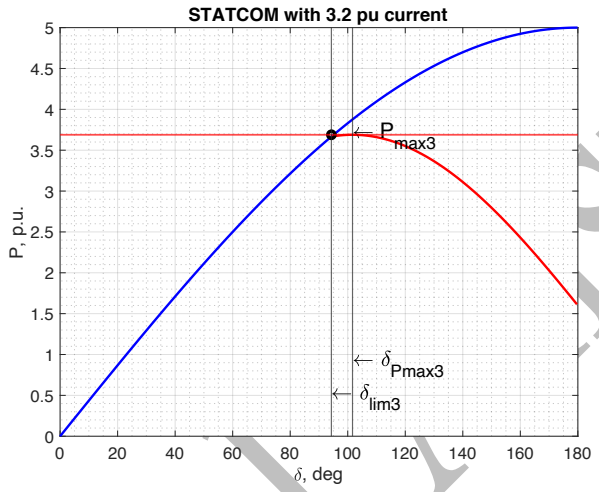
Ignoring the other solution 284.67°

[1 mark]

As δ_{Pmax} is larger than δ_{lim3} the maximum power is given by

$$P_{max3} = P_{max0} \sin\delta_{Pmax} + \frac{I_{max}}{2} \sin\frac{\delta_{Pmax}}{2} = 3.6889 \text{ p.u.}$$

[1 mark]



[1 mark]

Increase in steady-state stability limit

$\Delta P_{max3} = P_{max3} - P_{max0} = 1.1889$ p.u.

[1 mark]

b) Calculate the critical clearing angle (in degrees) with the SVC having 3.5 p.u. susceptance considering 1.6 p.u. of active power transfer through the line. Assume a self-clearing fault and zero power transfer during the fault.

[5]

Maximum power with ideal shunt reactive compensation

$$P_{max} = \frac{2V^2}{X} = 5 \text{ p.u.}$$

Initial power angle δ_0 is given by:

Model Answers

$$\delta_0 = 2 \times \sin^{-1}\left(\frac{P}{P_{max}}\right) = 37.326°$$

[1 mark]

After the reactive capacity of the SVC is reached i.e. for $\delta > \delta_{lim}$ the power angle characteristics is given by

$$P = \frac{4V^2}{X(4 - XB_{max2})}\sin\delta = P_{max-lim2}\sin\delta$$

$$P_{max-lim2} = \frac{4}{0.4(4 - 0.4 \times 3.5)} = 3.8462 \text{ p.u.}$$

[1 mark]

$$\delta_{max} = \pi - \sin^{-1}\left(\frac{P}{P_{max-lim2}}\right) = 155.42°$$

[1 mark]

Equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}} P\, d\delta = \int_{\delta_{cr}}^{\delta_{lim2}} \left(P_{max}\sin\frac{\delta}{2} - P\right) d\delta + \int_{\delta_{lim2}}^{\delta_{max}} (P_{max-lim2}\sin\delta - P)\, d\delta$$

[1 mark]

$$\delta_{cr} = 2 \times \cos^{-1}\left[\frac{P(\delta_{max} - \delta_0) + 2P_{max}\cos\frac{\delta_{lim2}}{2} - P_{max-lim}(\cos\delta_{lim2} - \cos\delta_{max})}{2P_{max}}\right]$$

$$= 92.798°$$

[1 mark]

Model Answers

Question 2 on System Operation with Low Inertia

2. a) To maintain the frequency stability in the future low-inertia system, it is important to meet the system frequency requirements on Rate of Change of Frequency (RoCoF), Frequency Nadir and Steady-state Frequency

i) Analytically derive the RoCoF constraint in terms of H_i (inertia constant of individual generators) and RoCoFmax from the swing equation.

[2]

$$2H\frac{\partial \Delta f(t)}{\partial t} + D \cdot \Delta f(t) = \Delta P_m(t) - \Delta P_L$$

[1 mark]

$$H = \frac{1}{f_0} \cdot \sum_{i=1}^{n} H_i \cdot P_i^{max} \geq H_{min} = -\frac{\Delta P_L}{2RoCoF_{max}}$$

[1 mark]

ii) Due to the challenge on measuring the instantaneous RoCoF, the average RoCoF over 500ms has been often used to trigger protection. For the same level of RoCoF limit (e.g. 0.5 Hz/s) and the same largest generation loss, discuss how instant RoCoF and average RoCoF limits would impact the required level of system inertia differently?

[2]

As the RoCoF keeps declining according to swing equation [1 mark], using average RoCoF over a certain period as the limit will lead to reduced requirement on system inertia [1 mark]

iii) The constraint that guarantees the system frequency nadir within a maximum admissible frequency deviation is given below:

$$\left(\frac{H}{f_0} - \mathrm{EFR} \cdot \frac{\mathrm{T_{EFR}}}{4 \cdot \Delta f_{\max}}\right) \cdot R \geq \frac{(P_{\mathrm{L}} - \mathrm{EFR})^2 \cdot \mathrm{T_d}}{4 \cdot \Delta f_{\max}}$$

Model Answers

Nomenclature

H: system level of inertia (Units: MW · s).
R: amount of frequency response scheduled (Units: MW).
EFR: amount of EFR scheduled (Units: MW).
P_L: value of the generation outage we want to secure against (Units: MW).
f_0: nominal electric frequency of the power grid (Units: Hz).
Δf_{max}: maximum admissible frequency deviation to avoid load shedding (Units: Hz).
T_d: Delivery-time of frequency response (Units: s).
T_{EFR}: Delivery-time of EFR (Units: s).

Other assumptions:

1. The largest generation unit is 1.8 GW and with reliability of 99%
2. Other numerical values: f_0 = 50 Hz, Δf_{max} = 0.8 Hz, T_d = 10 s, T_{EFR} = 0.5 s, EFR = 200 MW

The forecasted system inertia is following the distribution below:

Inertia (H)	Probability
140 GWs	5%
150 GWs	20%
160 GWs	50%
170 GWs	20%
180 GWs	5%

Draw the probability distribution of not meeting the frequency nadir requirement at various system inertia levels, assuming the frequency response is calculated and scheduled based on such level of inertia. To maintain the probability of meeting the frequency nadir requirement is above 99.9%, calculate how much frequency response (R) the system operator needs to procure.

[4]

Inertia (H)	Probability of not meeting naidr
140 GWs	0%
150 GWs	0.05%
160 GWs	0.25%
170 GWs	0.75%
180 GWs	0.95%

[2 marks]

To meet the reliability requirement of 99.9%, the frequency nadir has to be secure against inertia level of 150 GWs
[1 mark]

R=2695MW, EFR=200MW,
[1 mark]

Model Answers

iv) In addition to frequency response (R) with delivery time T_d and enhanced frequency response (EFR) with delivery time T_{EFR}, assume there is a new service (INS) that can be delivered instantly. Update the system frequency nadir requirement in (iii) to include this new service

[2]

Instant delivery of frequency response will be equivalent to reduce the last generation loss. Hence, the new constraint keeps the form below by now $P_L = P_{L-Org} - INS$

$$\left(\frac{H}{f_0} - \mathrm{EFR} \cdot \frac{\mathrm{T_{EFR}}}{4 \cdot \Delta f_{\max}}\right) \cdot R \geq \frac{(P_{\mathrm{L}} - \mathrm{EFR})^2 \cdot \mathrm{T_d}}{4 \cdot \Delta f_{\max}}$$

[2 marks]

b) Answer the following questions related to the implementation of frequency support services in renewable generation systems:

i) A wind turbine with a power rating of 12 MW operates above rated wind speed with a deloading of 1 MW (ie it generates 11 MW). Calculate the highest synthetic inertia constant, *H*, the wind turbine could emulate under this operating point if the worst RoCoF expected is of -2 Hz/s and the system's nominal frequency is 50 Hz.

[3]

The equation of the synthetic inertia is based on the swing equation:

$$\Delta P = -\frac{2H}{f_{nominal}} S_{base} \frac{df}{dt}$$

This can be used to calculate how much the power output of the generator would change for a given RoCoF. A positive RoCoF would cause the power to reduce, which in terms of its implementation in a wind turbine it would imply a reduction of the power output, which can be achieved by using speed and blade pitch angle control. A negative RoCoF would cause the power to increase, which requires the generator and its power converter to be sized for this increment. In this problem we know that our converter can handle an increase of power output of 1 MW, we can use this figure to calculate the highest *H* it would be able to emulate:

$$\mathrm{H} = -\frac{2\,\Delta \mathrm{P}\,\mathrm{f_{nominal}}}{S_{base}\frac{df}{dt}} = -\frac{2 \cdot 1\ \mathrm{MW} \cdot 50\ \mathrm{Hz}}{12\ \mathrm{MW} \cdot (-2\ \mathrm{Hz/s})} = 4.17\ s$$

ii) Which of the following forms of frequency support service from renewable generation would be more effective at reducing the worst RoCoF at the beginning of a frequency excursion caused by the loss of a large generator unit: synthetic inertia or automatic generation control (AGC) droop emulation? Justify your answer

[4]

BOOKWORK. The power order for these two services is of the form:

Model Answers

$$P = P_{prefault} + \Delta P$$

where

$\Delta P = -K_{inertia} \frac{df}{dt}$ in the case of synthetic inertia

and

$\Delta P = -K_{droop}(f - f_{rated})$ in the case of AGC droop emulation.

Looking at a typical frequency excursion for a large loss of generation comparing *df/dt* and *f*, it is clear that df/dt is large at the inception of the event, whereas the error of f builds up over time. Consequently, synthetic inertia is expected to have a greater weight at the inception of the event, whereas the AGC will become noticeable afterwards. This could be used to argue that synthetic inertia will have a greater effect on the initial RoCoF. On the other hand, the answer to the question of which of the two would have a greater effect also depends on how their settings compare. If K_{droop} is set to be very high, it is possible the the output of the emulated AGC becomes very high with a small error in *f*, which could make this controller have the same effect as a synthetic inertia controller.

Both answers can be considered to be correct as long as a justification is provided in the answer. An argument to defend that the AGC emulation is better thant he synthetic inertia would be harder to make.

iii) A wind turbine manufacturer suggests that their wind turbine would be able to boost its output power by up to 0.3 pu for a short period of time (e.g. 10 s) when operating at point A of its power curve (see Figure X). Is it possible? Justify your answer.

[3]

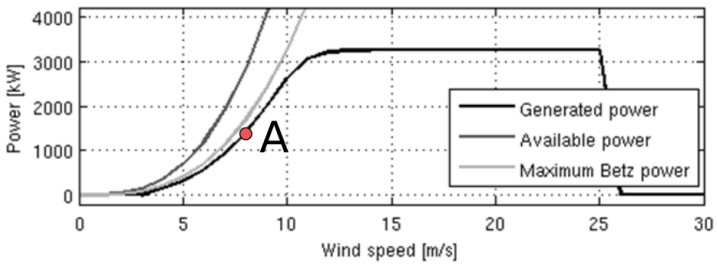


Figure X – Power curve of a 3.6 MW wind turbine.

BOOKWORK. Point A is within the MPPT operating range of the wind turbine, where the power that can be extracted from the wind is below the maximum power rating of the wind turbine. In this case, the turbine is at about 30% of the rated power. An increase of the power output of the turbine of +0.3 pu would move the power output to 60% of rated power, which is likely within the operating range of the power electronic converter of the wind turbine and its generator. If this happened, the torque the turbine would receive from the generator would be greater than the torque exerted by the wind; this would cause the wind turbine to decelerate and move away from the maximum power point (MPP). While this wouldn't be sustainable for a long period of time; it is sustainable for a short period of time such as a 10 or 20 seconds period. After this period, the turbine would have to go through a recovery process where the turbine's output power would be lower and the turbine would gain speed to go back to operating point A.

Model Answers

Question 3 on Power System State Estimation

(i) Why do system operators need a state estimator? [3]

An operator needs to know:

- To what extent network is loaded
- If there any equality or inequality constraint violation
- If any overloads following planned or unplanned contingencies
- How best to control and reconfigure the network to utilise any spare capacity

(ii) What are the fundamental differences between state estimation in transmission and distribution network operation? [5]

Transmission System

- More measurements
  (Enough to guarantee observability)
- Methodologies and tools already exist
  (e.g. WLS, M-estimators, LAV)
- Pseudo measurements can be modelled as zero injection or inequality constraints
- Loads can be modelled as constant impedance type
- Limited variation in network parameters

Distribution System

- Less / inadequate measurements (observability not guaranteed)
- Lack of rigorous methodologies and tools for handling restricted measurements
- No definite model for pseudo measurements
  (Modelling pseudo measurements is challenging in DNO)
- High variability in network parameters (due to thermal variations etc)
- Loadings are unbalanced in nature

(iii) In the context of power transmission state estimation starting with measurements and states, formulate weighted least square (WLS) as maximum likelihood estimate (MLE). [5]

Set up:
State vector: node voltage and angle

Model Answers

Measurement: line real and reactive power flow, line current, bus voltage, power injection

Relate the measurement as function of state $z = h(x) + \varepsilon$

Error characterization: Gaussian, independent (uncorrelated), null average (unbiased) , standard deviation σ : $N(0, \sigma)$

$\text{cov}(\varepsilon) = \text{diag}(\sigma_i^2) = R$

Minimise a scalar function : $J(x) = [z - h(x)]^T W [z - h(x)]$;

$$J(x) = \sum_i W_i [z_i - h_i(x)]^2$$

Weight matrix : $W = R^{-1}$ Residual vector : $r = z - h(x)$

Solution:

First derivative = 0

$H^T W [z - h(x)] = 0$, $H = \partial h / \partial x$

Newton's approach

$G(x^k)\Delta x = H^t(x^k) W [z - h(x^k)]$

$G(x^k) = H^t(x^k) W H(x^k)$

G-gain matrix.

Obtain Δx and then update x^k. continue the iterative process until convergence

(iv) What is equality constrained state estimation? [3]

When the objective function of MLE has some constraints in the formulation we have equality constraints state estimation. Zero injection is one such measurement which is 100% correct, so best is to remove it from the measurement set rather than assigning a high weight. It reduces the dimensionality of the normal equation and along with constraint the solution is sought through Lagrange function. This is known as constraints state estimation. The problem set up is well conditioned for solution but symmetry is lost.

(v) Under what circumstances constrained state estimation is adopted? [2]

When all measurements contributing to normal equation are of not equal quality, the normal equation is not well conditioned. In distribution system there are very few high-quality measurement, many pseudo measurements with large covariance, some leverage measurement – all needing highly varying weights. The normal equation is ill conditioned, requiring robust factorisation etc. The solution sometimes does not converge. Under such situation it is prudent to go for constraints state estimation approach

Model Answers

(vi) What is the role of critical measurement in state estimation? [2]

Critical measurement is the one that guarantees solvability and unique solution. It is also linked to observability. If critical measurement is lost, the whole power system is split into two sub systems and unified solution is not possible. The system is unobservable with the remaining set of measurements The critical measurement is also considered not to have residual error.

[20 marks]

Model Answers

Question 4 on Integrated Energy Systems

Considering the electrification of heat sector, heat appliances through thermal energy storage (TES) could provide flexibility to the electricity system and support more cost-effective integration of variable wind generation. Consider a system with capacity of wind generation of 35GW, nuclear generation 12.4GW, with the rest of the load being supplied by CCGT plants, which are all the same type with same characteristics. Electricity demand includes non-heat demand and electrified heat demand. Operating reserve is required to deal with uncertainties in wind generation production, which is managed by part-loading CCGT.

Key characteristics of the CCGT plant are:
Capacity: 600 MW
Minimum Stable Generation (MSG): 300 MW
Minimum online time: 2 hours (once a CCGT unit is started up, it would be running for at least 2 hours)

System operational conditions and reserve requirements are given in the table below, considering 2 consecutive hours of system operation.

	1st hour	**2nd hour**
Wind available (MW)	14,600	28,800
Nuclear (MW)	12,400	12,400
Electricity Demand (MW) - Non-heat demand - Electrified heat demand	51,000 36,000 15,000	51,000 36,000 15,000
Reserve requirement (MW)	9,600	13,500

a) Assuming that
- TES is not available,
- 6 CCGT units have already been online for more than 1 hour at the beginning of the 1st hour,

answer the following questions:

i) how many extra CCGT units should be started up in the 1st hour? [4.5]

50 CCGTs

In the 1st hour,
Power from CCGT = 51000MW - 14600MW - 12400MW= 24000MW
Reserve from CCGT = 9600MW
Number of CCGT = (24000 + 9600)/600 = 56
Therefore, 56-6=50 CCGTs should be started in the 1st hour.

ii) Calculate wind curtailment in the 2nd hour. [4.5]

5200MW

In the 2nd hour, considering the minimum online time, at least 50 CCGT must be online.

Model Answers

Reserve from CCGT= 13500
Number of CCGT required = 13500/300=45<50
Minimum power from CCGT = 50*300=15000MW
Wind required = 51000-15000-12400 = 23600MW
Wind curtailment = 28800-23600 = 5200MW

b) TES can accommodate surplus of wind power and supply heat demand.
Assume that
- 3500MW (power equivalent) TES is available,
- The initial thermal energy stored in TES is 3000MWh (power equivalent),
- Heat loss of TES is ignored.
- 6 CCGT units have already been online for more than 1 hour at the beginning of the 1st hour,

and answer the following questions:

i) calculate the minimum wind curtailment in the 2nd hour.

[6]

200MW

In the 1st hour,
Power from CCGT = 51000MW-14600MW-12400MW-3000MW= 21000MW
Reserve from CCGT = 9600MW
Number of CCGT = (21000 + 9600)/600 = 51
Therefore, 51-6=45 CCGTs should be started in the 1st hour.

In the 2nd hour, considering the minimum online time, at 45 CCGT must be online.
Reserve from CCGT= 13500
Number of CCGT required = 13500/300=45
Minimum power from CCGT = 45*300=13500MW
Wind required = 51000-13500-12400 = 25100MW
3500MW wind power can be stored in TES
Wind curtailment = 28800-25100-3500 = 200MW

ii) If CO_2 emission associated with the operation of CCGT is 0.4ton/MWh, how much CO_2 is saved by using TES?

[5]

1800ton.

In the 1st hour,
24000MW-21000MW=3000MW less power is generated by CCGT.
In the 2nd hour,
15000MW-13500MW=1500MW less power is generated by CCGT.
(3000+1500)*0.4=1800ton CO2 is saved by using TES.

Model Answers